**Table 3** Overview of included studies on health statue utility values

| Region | Study Author year | Study design/study type | Patient population | Treatment | Follow-up times assessed | Utility assessment method | Scale | Baseline mean (SD) | Mean change at last follow-up (SD) |
|---|---|---|---|---|---|---|---|---|---|
| **Utilities** | | | | | | | | | |
| *EQ-5D* | | | | | | | | | |
| Global | CheckMate 436 Zinzani 2019 [45] | Phase 1/2 single-arm study | Adult patients with R/R PMBCL | Nivolumab + brentuximab vedotin | Baseline<br>Cycle 5<br>Follow-up 1 | EQ-5D-3L mobility | 0 (no problems) to 3 (extreme problems) | Some problems: 14%<br>Extreme problems: 0 | Some problems: 21%<br>Extreme problems: 0 |
| | | | | | | EQ-5D-3L self-care | 0 (no problems) to 3 (extreme problems) | Some problems: 11%<br>Extreme problems: 0 | Some problems: 14%<br>Extreme problems: 0 |
| | | | | | | EQ-5D-3L activity | 0 (no problems) to 3 (extreme problems) | Some problems: 39%<br>Extreme problems: 0 | Some problems: 21%<br>Extreme problems: 0 |
| | | | | | | EQ-5D-3L pain | 0 (no problems) to 3 (extreme problems) | Some problems: 71%<br>Extreme problems: 7% | Some problems: 21%<br>Extreme problems: 7% |
| | | | | | | EQ-5D-3L anxiety | 0 (no problems) to 3 (extreme problems) | Some problems: 46%<br>Extreme problems: 0 | Some problems: 57%<br>Extreme problems: 0 |
| Global | SADAL<br>Shah 2021 [38]<br>Casasnovas 2020 [69] | Phase 2b single-arm trial | Adult patients with R/R DLBCL with 2 + prior lines of therapy | Selinexor | Baseline<br>Cycle 2<br>Cycle 3<br>Cycle 4<br>Cycle 5<br>Cycle 6<br>Cycle 7 | EQ-5D-5L health index score | 0 (dead) to 1 (perfect health) | Responders: 0.789 (95% CI, 0.767, 0.811)<br>Nonresponders: 0.801 (95% CI, 0.781, 0.822) | Responders: − 0.010 (95% CI, − 0.15, − 0.05)<br>Nonresponders: − 0.274 (NR) |
| Global | ZUMA-1<br>Lin 2019 [42]<br>ClinicalTrials.gov [70] | Phase 2 single-arm trial | Adult patients with 3L + DLBCL of therapy (Cohorts 3−6) | Axi-cel | Baseline<br>Week 4<br>Month 3<br>Month 6 | EQ-5D-5L visual analog scale | 0 to 100 (higher scores reflecting better health) | Cohort 3: 71.2 (21.3)<br>Cohort 4: 69.5 (18.8)<br>Cohort 5: 66.7 (20.7)<br>Cohort 6: 70.9 (17.0) | Cohort 3: + 5.9[a]<br>Cohort 4: + 15.6[a]<br>Cohort 5: + 10.4[a]<br>Cohort 6: + 8.9[a] |
| | | | Adult patients with 3L + LBCL (safety management study) | | Baseline<br>1 month<br>2 months<br>3 months<br>6 months | EQ-5D-5L health index score | 0 (dead) to 1 (perfect health) | 0.80 (0.17) | 0.82 (0.21) |
| US | TRANSCEND NHL 001<br>Patrick 2021 [17]<br>Patrick 2020 [37]<br>Patrick 2019 [35]<br>Patrick 2019 [36] | Phase 1 single-arm trial | Adult patients with R/R DLBCL with 2 + prior lines of therapy | Liso-cel | Baseline<br>1 month<br>2 months<br>3 months<br>6 months<br>9 months<br>12 months<br>18 months | EQ-5D-5L health index score | 0 (dead) to 1 (perfect health)[a] | 0.817 (0.120) | + 0.061[b] (0.013) |
| | | | | | | EQ-5D-5L visual analog scale | 0 to 100 (higher scores reflecting better health) | 68.3 (19.5) | + 10.4 (5.4) |
| NR | Orfanos 2022 [46] | Phase 2 single-arm trial | 2L and heavily pretreated patients with R/R DLBCL | Naratuximab emtansine + rituximab | Baseline<br>End of therapy | EQ-5D (unspecified) | 0 (dead) to 1 (perfect health) | Responders: 0.78 (NR)<br>Nonresponders: 0.73 (NR) | Responders: 0.77 (NR)<br>Nonresponders: 0.67 (NR) |

**Table 3** (continued)

| Region | Study Author year | Study design/study type | Patient population | Treatment | Follow-up times assessed | Utility assessment method | Scale | Baseline mean (SD) | Mean change at last follow-up (SD) |
|---|---|---|---|---|---|---|---|---|---|
| UK | Howell 2022 [44] Howell 2020 [43] | Vignette-based study | General adult population in UK | Hypothetical CAR T cell therapy pathway | NA | EQ-5D-5L health index score | 0 (dead) to 1 (perfect health)[c] | 0.86 (0.17) | NR |
| | | | | | | EQ-5D-5L visual analog scale | 0 to 100 (higher scores reflecting better health) | 82.15 (13.54) | NR |
| *TTO* | | | | | | | | | |
| UK | Howell 2022 [44] Howell 2020 [43] | Vignette-based study | General adult population in UK | Hypothetical CAR T cell therapy pathway | NA | TTO (1-year time horizon) | 0 (dead) to 1 (perfect health)[c] | NR (results are disutilities of a hypothetical treatment pathway [i.e., no baseline]) | CAR T cell therapy for LBCL with grade 1 CRS: − 0.01 (0.04)[d]<br>CAR T cell therapy for LBCL with grade 2 CRS: − 0.05 (0.09)[d]<br>CAR T cell therapy for LBCL with grade 3/4 CRS: − 0.23 (0.24)[d]<br>CAR T cell therapy for LBCL with grade 1/2 NEs: − 0.04 (0.07)[d]<br>CAR T cell therapy with grade 3/4 NEs: − 0.18 (0.22)[d] |

*2L* second line; *3L+* third line or later; *AE* adverse event; *axi-cel* axicabtagene ciloleucel; *CAR* chimeric antigen receptor; *CI* confidence interval; *CRS* cytokine release syndrome; *DLBCL* diffuse large B cell lymphoma; *EQ-5D-3L* EuroQol 5-dimensional, 3-level; *EQ-5D-5L* EuroQol 5-dimensional 5-level; *LBCL* large B cell lymphoma; *liso-cel* lisocabtagene maraleucel; *NA* not available; *NE* neurological event; *NR* not reported; *PMBCL* primary mediastinal large B cell lymphoma; *R/R* relapsed or refractory; *SD* standard deviation; *TTO* time tradeoff; *UK* United Kingdom; *US* United States

[a]EQ-5D visual analog scale values are calculated using the 6-month end score

[b]Mean change was calculated based on baseline and follow-up values

[c]Utility values can occasionally be below zero, reflecting a health state less preferable than dead

[d]Values represent the difference between the utility of the health state without an AE (i.e., base LBCL health state) and with an AE